

Task 19: Git Repository and Basic Operations

AIM

To create a Git repository for a sample project and demonstrate basic Git operations such as initializing a repository, staging files, committing changes, maintaining version history, and pushing to GitHub.

ALGORITHM

- Step 1: Initialize a new Git repository using git init.
- Step 2: Create sample project files (e.g., index.html, style.css).
- Step 3: Stage files using git add.
- Step 4: Commit changes with a descriptive message using git commit.
- Step 5: View version history using git log.
- Step 6: Create a remote repository on GitHub.
- Step 7: Link local repo to remote using git remote add origin.
- Step 8: Push the repository to GitHub using git push.

OUTPUT

Terminal Commands and Output:

Step 1: Initialize Repository

```
$ mkdir sample-project && cd sample-project
$ git init
Initialized empty Git repository in /home/user/sample-project/.git/
```

Step 2: Create Files

```
$ echo "<html><body><h1>Hello World</h1></body></html>" > index.html
$ echo "body { font-family: Arial; }" > style.css
$ echo "# Sample Project" > README.md
```

Step 3: Check Status and Stage Files

```
$ git status
Untracked files: index.html, style.css, README.md
$ git add .
$ git status
```

Changes to be committed: new file: index.html, style.css, README.md

Step 4: Commit Changes

```
$ git commit -m "Initial commit: Add project files"
[main (root-commit) a1b2c3d] Initial commit: Add project files
3 files changed, 5 insertions(+)
```

Step 5: Make Another Change and Commit

```
$ echo "<p>Welcome to my site</p>" >> index.html
$ git add index.html
$ git commit -m "Add welcome paragraph to index.html"
[main e4f5g6h] Add welcome paragraph to index.html
```

Step 6: View Version History

```
$ git log --oneline
e4f5g6h Add welcome paragraph to index.html
a1b2c3d Initial commit: Add project files
```

Step 7-8: Push to GitHub

```
$ git remote add origin https://github.com/username/sample-project.git
$ git branch -M main
$ git push -u origin main
Enumerating objects: 8, done. Writing objects: 100% (8/8), done.
To https://github.com/username/sample-project.git
* [new branch] main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
```

RESULT

Successfully created a Git repository, demonstrated staging, committing, viewing history, and pushed the project to a remote GitHub repository. Version history is maintained with meaningful commit messages.